

Leyna May-Gurr

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SUMMARY

Looking for software engineering roles. I bring a Mathematics PhD and experience building software and turning complex logic into clean, functional code. Strong foundation in Python and JavaScript, developed through publishing recursive algorithms and building full-stack projects including 3D physics simulations and browser games.

I have professional experience teaching Python and R to undergraduate engineering students and marking technical assessments at the University of Sheffield. A strong collaborator and communicator with experience running seminar series, organising international conferences, and volunteering for Women in Tech workshops.

Based in Munich and eligible to work in Germany without visa sponsorship.

EDUCATION

University of Sheffield 2025

PhD in Mathematics. Thesis 'Hidden Symmetries, Trees and Operads'

Durham University 2020

Masters in Mathematics, First class honours. Thesis 'Morse Theory: Morse-Bott Functions and Homology'

EXPERIENCE

Graduate teacher 2021 – 2025

University of Sheffield

- Taught Python and R to undergraduate engineering students; led tutorials in mathematics and engineering
- Marked examinations and assessments, reviewing and debugging student code with constructive feedback
- Supervised a UKRI-funded undergraduate summer project on knot theory (June–July 2023)

PROJECTS

Noughts and Crosses Game | *Python, FastAPI, JavaScript, Vue.js, HTML, CSS* March 2026

- Built a real-time multiplayer game with a FastAPI backend handling WebSocket connections, game state management, and room creation/joining
- Redesigning the front-end in Vue.js, replacing a prototype with a more polished implementation
- Can be found at noughts-and-crosses.leynamay.com and on GitHub at github.com/LeynaM/noughts-and-crosses

Tilt Game | *JavaScript, Three.js, React, Node.js, SQLite, HTML, CSS* January 2025

- Developed a full-stack physics-based browser game using Three.js, implementing custom physics algorithms for ball movement, platform tilting, and collision detection
- Built mouse controls for platform interaction and used SQLite to store scores with a React leaderboard
- Deployed to Oracle Cloud using GitHub Actions
- Can be found at tilt.leynamay.com and on GitHub at github.com/LeynaM/tilt-game

PUBLICATIONS

Quartic graphs which are Bakry-Émery curvature sharp | *Python* March 2020

- Designed and implemented a recursive algorithm to generate unique graphs with certain curvature properties
- Published in Discrete Mathematics, Volume 343, Issue 3 (<https://doi.org/10.1016/j.disc.2019.111767>)

SKILLS

Frontend: JavaScript, HTML, CSS, Vue.js, React

Backend: Python, FastAPI, Node.js, SQLite

Tools: Git, Linux, LaTeX

Languages: English (Native), German (B1)